

Progress Report

- Increment 1 -

Group Metroidvania

1) Team Members

Skyler Haas, smh23a, 149013192

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2) Project Title and Description

Our project is currently titled “Metroidvania” and, as the name implies, will be a short Metroidvania-style video game in which players can control a character from a 2D side scrolling perspective to explore an interconnected world, with a focus on smooth movement and tight, responsive controls. By exploring the map and defeating enemies scattered around the world, players can collect various items and upgrades to help them traverse the world, unlock new pathways, and aid in combat with enemies and the final boss. The game will be designed to be beatable in one sitting.

3) Accomplishments and overall project status during this increment

Player movement and animations has been the main focus for this increment, where we consistently tweaked how the player would move and react with the world around them in order to have the most satisfying player experience. Compared to our initial scope, we have made significant progress as we initially intended for player movement to be a high priority from the beginning.

4) Challenges, changes in the plan and scope of the project and things that went wrong during this increment

Some ideas aren't possible due to our initial scope of the project as well as there not being much time for us to implement them all. One idea we couldn't implement was the idea for a grappling hook item that allowed for the player to grapple onto walls and attack enemies with it. We decided not to include the grappling hook due to how unnatural and rough it felt for a player to use, and because of this we decided to shift our scope of the project to something a little smaller.

5) Team Member Contribution for this increment

Skyler Haas: Responsible for constructing and narrating the video demonstrating the Progress Report with guidance/oversight from other members. Responsible for writing the "Non-functional requirements" section of the RD doc by drafting up the non-functional requirements of our project such as control responsiveness, level readability, and reliability. Provided oversight and input for the IT doc, primarily the "Execution-based Non-Functional Testing" section as it related heavily to his work on the RD doc. Contributions to the source code include designing some basic enemy behavior and prototypical level design for the introductory/tutorial area of the game.

Raul Medrano: Responsible for writing the first four parts of the Progress Report (listing the team members, providing a basic description of the project, describing accomplishments and project status for this increment, and describing the challenges and scope changes during this increment). Responsible for writing the "Functional requirements" section of the RD doc by drafting up and listing the functional requirements of our project such as the interconnected map and player moveset requirements. Provided oversight and input for the

IT doc, primarily the "Execution-based Functional Testing" section as it related heavily to his work on the RD doc. Contributions to the source code include designing and implementing hand-drawn sprites and animations for the player character to replace the placeholders we previously had, as well as experimenting with other graphical elements such as tilesets and parallax scrolling for the background. Provided some input and oversight for the video, namely providing sprites and gameplay clips for Skyler to show off as demonstrations of the team's progress.

Armani Ruiz: Responsible for writing the "Team Member Contribution for this increment" section of the Progress Report, listing each team member and describing their contributions to each deliverable. Responsible for writing the "Operating Environment" and "Assumptions and Dependencies" sections of the RD doc, laying out the game engine and system requirements for our project as well as the assumptions and dependencies that inform and guide the development of the project overall. Responsible for constructing and organizing the IT doc using the inputs provided by other team members, with primary focus on individually developing the "Non-execution-based Testing" section. Contributions to the source code were admittedly minimal for the first increment (aside from ongoing work on enemy behavior and a save/load feature) as he instead primarily focused on establishing an organized Scrum-inspired workflow for the project, including taking the initiative to set up a sprint timeline to pace the team's work, encouraging team members to utilize the issue tracker on GitHub, and keeping record of every team member's tasks for each sprint. Provided some input and guidance on the video including helping to decide what to include on each slide.

Evan Treadwell: Responsible for writing the "Stakeholder communication" section of the Progress Report, drafting an email describing our progress that would be sent to the publishing team/company for our game if we hypothetically wanted to publish it. Also responsible for the "Class diagram and/or sequence diagram" section of the RD doc, designing an overview of how different aspects of our system's architecture, such as UI/HUD elements, enemies, and the player interact with each other within the first area of the game that the team has been developing in this first increment. Provided oversight and input for the IT doc, primarily the "Programming languages" section. Contributions to source code include ongoing development of various UI and HUD elements for the game such as the pause menu and player health bar. Provided some input and guidance on the video including helping to decide what to include on each slide.

John Wibert: Responsible for writing the "Plans for the next increment" section of the Progress Report, describing some ideas for what elements the team will focus on going forward such as level design, UI elements, and improving code/file structuring. Responsible for the "Use Case Diagram" section of the RD doc by designing a simple use case diagram to show how a player may interact with the game's elements and systems on a surface level. Provided oversight and input for the IT doc, namely the "Platforms, APIs, Databases, and other technologies used" section. Contributions to the source code include developing the player character's movement feel and controls, primarily by conceptualizing and implementing the various unique traversal/combat abilities that the player will unlock throughout the game such as the double jump, air attack, and dash. Worked closely with Skyler to develop an outline for the video.

6) Plans for the next increment

In the next increment, we plan to make major progress in a few main areas. The first and most important is the level design; we plan to have rough versions of each major zone the game is going to be split into completed by the end of this increment. Next is further improvements to UI elements and behind-the-scenes elements, such as better code and file structuring as well as beginning to develop systems for saving and continuing the game. Finally, we want to continue working on the enemies and general design of the challenges the player will face while also fine tuning the player's movement, hitboxes, etc.

7) Stakeholder Communication

Hi Publisher team,

Metroidvania has reached an early playable state. The player's core movement set is complete, including running, jumping, double jumping, and an air dash, along with melee combat and an early iteration of an mid-

air bounce attack that will provide movement depth in combat and platforming. Players can now fight two distinct enemy types in a test area: a slow, heavy hitter and a fast attacker. A health display and pause menu are also in place.

Initial player art has been completed, and map art is currently in the works.

Our next milestone focuses on drafting level design, connected rooms and ability-based progression, which will bring the game's exploration structure online.

We're happy to share a playable build at your convenience.

8) Link to video

<https://1drv.ms/p/c/cd75b58f6b4ed0f7/1QAJbPuQjwkwQZUVwP0BGaeAAfD95-txtr0iyaKjS-vFEjI?e=5pYNfD>